

Abeer Banerjee

PhD Researcher | Computational Imaging

abebanerjee.github.io

Munich, Germany

abebanerjee.io@gmail.com 

[LinkedIn Profile](#) 

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Professional Summary

PhD researcher in computational imaging, working on physics-informed and learned reconstruction methods for ill-posed inverse problems in biomedical and optical imaging. Published research on generative and physics-based models for image restoration, event-based vision, and 3D reconstruction. Currently based in Munich (DAAD fellow, TUM/Helmholtz Munich).

Highlights

Affiliations	DAAD PhD Fellow at Technical University of Munich (TUM) — Guest Researcher at Helmholtz Munich — PhD Candidate at AcSIR
Experience	5 years of experience in Computational Imaging, Machine Learning, Inverse Problems, Computer Vision, Neuromorphic Vision, Biomedical Imaging
Outputs	12+ peer-reviewed publications [IEEE Transactions, Elsevier, Springer, BMVC, ICIP, IJCNN] — 2 registered copyrights — 1 research grant

Research Experience

Research Theme: Physics-informed Computational Imaging

CSIR–Central Electronics Engineering Research Institute (CEERI), India [2020 - 2022]

- Junior Research Fellow - Master's thesis on generative grayscale image colorization - Foundation work on inverse imaging - Published at TVCJ.

CSIR–Central Electronics Engineering Research Institute (CEERI), India [2022 - 2024]

- Senior Research Fellow - Developed physics-informed image reconstruction algorithms - Reproducible experiment-tracking pipelines - Trained GAN-based restoration models on DGX-1 HPC infrastructure - Research focus on lensless imaging.
- Published research at international conferences and high-impact journals - Mentored junior lab members - Contributed in grant writing - Teaching assistance.

Technical University of Munich (TUM), Germany

Institute of Biological and Medical Imaging (IBMI), Helmholtz Munich, Germany [2024 - Present]

- DAAD PhD Sandwich Fellow - Developed physics-informed learning frameworks with implicit neural representations for 3D vasculature reconstruction from optoacoustic microscopy images.
- Poster presented at IEEE ISBI 2026, London. Extended manuscript in preparation.

Research Theme: Neuromorphic Vision

CSIR–Central Electronics Engineering Research Institute (CEERI), India [2022 - 2024]

- Senior Research Fellow - Developed deep learning methods for eye-gaze estimation under extreme low-light using custom event-based (DVS) camera headgear.
- Contributed to face anti-spoofing, human activity recognition and privacy-preserving vision systems. Published research presented at IEEE IJCNN, ICVGIP, IHCI, and other international conferences.

Selected Publications

Journal Articles - First Author

- TETCI: *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2025, IF: 6.5. [[Paper](#)]
- EAAI: *Engineering Applications of Artificial Intelligence*, Elsevier, 2025, IF: 9.0. [[Paper](#)]
- TVCJ: *The Visual Computer Journal*, Springer, 2023, IF: 3.4. [[Paper](#)]

Peer-reviewed Conference - First Author

- BMVC 2023: *British Machine Vision Conference*, h5-index: 77. [[Paper](#)]
- ICIP 2023: *IEEE International Conference on Image Processing*, h5-index: 61. [[Paper](#)]
- IJCNN 2024: *IEEE International Joint Conference on Neural Networks*, h5-index: 60. [[Paper](#)]

Complete publication list: [Google Scholar Profile](#)

Education

Integrated PhD in Computational Imaging 2020–Present

Academy of Scientific and Innovative Research (AcSIR), India [Anabin H+]

Advisors: Dr. Sanjay Singh, Prof. Vasilis Ntziachristos

Master's Thesis: Generative Colorization of Grayscale Images [2022]

CGPA: 9.13/10

B.Tech. Electronics and Communication Engineering 2016–2020

University of Calcutta, India [Anabin H+]

CGPA: 8.64/10

Core Expertise

Research Methods	Physics-informed Learning, Generative Models, Learned Optimization, PINNs, Implicit Neural Representations, GANs, Representation Learning, Diffusion Models, Spiking Neural Networks
Scientific Computing	Python (advanced), MATLAB, PyTorch (preferred), TensorFlow, OpenCV, Docker, Git, Linux, TensorBoard, L ^A T _E X
Languages	English (fluent), Bengali (native), Hindi (int.), German (basic)

Academic Service

- **Invited Reviewer:** Neurocomputing, Expert Systems with Applications, SIVP, Multimedia Systems, Computers & Electrical Engineering, IEEE Transactions on II, Scientific Reports, The Visual Computer, IET Computer Vision, ACML, IEEE IJCNN, ICVGIP, and other international venues.
- **Session Chair:** IEEE International Joint Conference on Neural Networks (IJCNN), 2024

Honours & Awards

- DAAD Bi-national Doctoral Degree Fellowship at Technical University of Munich (2024–2026)
- Junior and Senior Research Fellowship, GATE 2020, Council for Scientific and Industrial Research (CSIR), India (2020–2025)